1

This functions quite similar to a logistic regression

So assumptions are:

- $0 \leq C_i(x) \leq 1$ and is fixed
- n samples with $\forall y^{(i)} \in y, y^{(i)} \in \{0, 1\}$
- bias fixed at 0.5
- We will use a loss function of logistic loss
- $C(x^{(i)})$ returns a list of probabilities corresponding such that the jth probability is $C_j(x^{(i)})$ or the probability that the jth model predicts for our input $x^{(i)}$
- $\sigma(z)$ same as logistic regression so $\sigma(z) = 1$ iff $z \geq 0$ else 0
- η and epoch numbers are given to us

Algorithm pseudocode:

```
m = # Models
n = # Data Points

w = [None] * m

for w_j in w:
  w_j = 1/m

p = [None] * n

for p_i in p:
  p_i = C(x^{(i)})

for i in range(NUM_EPOCHS):
  z = w^T p - 0.5
  y_pred = sigma(z)
  dw = dotproduct(x, y_pred-y)
  w -= dw * eta

return w
```

2

We find that for correct classifications, we multiply by $1/\sqrt{3}$ and for incorrect classigications we multiply by $\sqrt{3}$. Thus $e^{-\alpha} = 1/\sqrt{3}$

This means that $\alpha = \ln\sqrt{3}$

We know by the definition of α that:

$$\alpha = 0.5\ln(\tfrac{1-\epsilon}{\epsilon})$$

So we get the following:

$$\ln\sqrt{3} = \alpha = 0.5\ln(\tfrac{1-\epsilon}{\epsilon})$$

$$\rightarrow \ln\sqrt{3} = 0.5\ln(\tfrac{1-\epsilon}{\epsilon})$$

$$\rightarrow 0.5\ln 3 = 0.5\ln(\tfrac{1-\epsilon}{\epsilon})$$

$$\rightarrow 3 = \tfrac{1-\epsilon}{\epsilon}$$

$$\rightarrow 3\epsilon = 1-\epsilon$$

$$\rightarrow \epsilon = \tfrac{1}{4}$$

3

We are given the initial following features:

- $x_1^{(k)} = k \times 10^{-1}$
- $x_2^{(k)} = k$
- $x_3^{(k)} = k \times 10$

Thus our degree 2 polynomial transformation will result in the following features:

- $x_0^{(k)} = 1$
- $x_1^{(k)} = k \times 10^{-1}$
- $x_2^{(k)} = k$
- $x_3^{(k)} = k \times 10$
- $x_4^{(k)} = (x_1^{(k)})^2 = k^2 \times 10^{-2}$
- $x_5^{(k)} = (x_2^{(k)})^2 = k^2$
- $x_6^{(k)} = (x_3^{(k)})^2 = k^2 \times 10^2$
- $x_7^{(k)} = x_1^{(k)} x_2^{(k)} = k^2 \times 10^{-1}$
- $x_8^{(k)} = x_1^{(k)} x_3^{(k)} = k^2 \times 10^1$
- $x_9^{(k)} = x_2^{(k)} x_3^{(k)} = k^2$

4

Given our initial centroids are $c_1 = (2, 0)$ and $c_2 = (-2, 0)$ with the below initial points:

- $x^{(1)} = (-1, 0)$
- $x^{(2)} = (1.5, 0)$
- $x^{(3)} = (-1, -1)$
- $x^{(4)} = (1, 1)$
- $x^{(5)} = (2, 1)$

After this iteration we find that $x^{(1)}, x^{(3)}$ are closest to c_2 and that $x^{(2)}, x^{(4)}, x^{(5)}$ are closest to c_1

And membership is for cluster 1:

- $x^{(2)}, x^{(4)}, x^{(5)}$

Thus membership is for cluster 2:

- $x^{(1)}, x^{(3)}$

Thus we find our new centroids are:

- $c_1 = ([1.5 + 1 + 2]/3, [0 + 1 + 1]/3) = (1.5, \frac{2}{3})$
- $c_2 = ([-1 - 1]/2, [0 - 1]/2) = (-1, 0.5)$

5

6

7

8

9

10